

Hackathon Problem Statements

Regd No: 25091A3224, 25091A3230

Problem Statement No: 85

Recycling Center Route Optimizer Build a route optimization tool using shortest-path algorithms to plan efficient collection routes for recyclable materials from multiple drop-off points to a central recycling facility, minimizing fuel and time costs. Use OOP to model collection points, vehicles, and recyclable material categories. This improves the economic viability and efficiency of community recycling programs.

Regd No: 25091A3267, 25091A32D4

Problem Statement No: 100

Investment Portfolio Diversification Optimizer Design a portfolio optimization tool using dynamic programming to allocate a fixed investment budget across asset categories (equity, debt, gold) to maximize expected returns within a given risk tolerance, based on historical return and volatility datasets. Use OOP to model asset classes and portfolio strategies. This introduces students to practical quantitative investment concepts through a working simulation.

Regd No: 25091A3238, 25091A32F8

Problem Statement No: 102

ATM Cash Replenishment Optimizer Create an ATM cash management system using demand forecasting on historical withdrawal patterns combined with priority scheduling to optimize replenishment routes and timing, minimizing both cash-out incidents and idle cash. Use OOP to model ATMs, cash-in-transit vehicles, and replenishment schedules. This reduces operational costs for banks while improving customer service reliability.

Regd No: 25091A3299, 25091A32B7

Problem Statement No: 106

Flood Evacuation Route Planner Build an evacuation route planning tool using shortest-path algorithms on a real or simulated terrain/road-network graph to compute the fastest safe evacuation routes from flood-prone zones to designated shelters, accounting for blocked or submerged roads. Use OOP to model roads, shelters, and hazard zones with dynamic states. This supports disaster management authorities in generating real-time evacuation guidance during flood emergencies.

Regd No: 25091A32B5, 25091A3277

Problem Statement No: 121

Public Grievance Categorization & Routing System Build a grievance management system using NLP-based text classification to automatically categorize citizen complaints (water, roads, sanitation, corruption) and route them to the correct department using a rule-based OOP dispatcher. Track resolution time to generate departmental accountability reports. This improves transparency and efficiency in government grievance redressal, addressing widespread citizen frustration with slow bureaucratic responses.

Regd No: 25091A32N5, 25091A32N4

Problem Statement No: 143

Network Intrusion Detection Simulator Build a network intrusion detection simulator using anomaly detection algorithms on simulated network traffic log datasets to flag unusual patterns indicative of potential intrusions (port scanning, unusual data volume). Use OOP to model network nodes, traffic logs, and alert rules. This gives students hands-on experience with core cybersecurity monitoring concepts using real data science techniques.